

Client Authentication Agent (CAA)

Installation Guide for Linux

PES University – RR Campus

Overview

The Client Authentication Agent (CAA) is a lightweight tool used to authenticate your system with the PES University network. Once configured, it allows seamless internet access without repeatedly logging into the captive portal.

Using CAA helps avoid issues such as maximum login limits and eliminates the need to manually log out and log back in when switching devices or reconnecting to the network.

1. Download the CAA Package

- Open your browser and navigate to: <https://rr.pes.edu:4443/>
- Login using your **Internet Captive Portal** credentials
- Go to the **Download** section
- Download the **Linux Authentication Client** (caa_x64.tar.gz)

2. Extract the CAA Package

```
tar -xvf caa_x64.tar.gz
cd caa*
```

Expected structure:

```
.caa/
caa.conf
ca-cert.pem
README
bin/
caa
```

3. Install Configuration Files

```
cp -r .caa ~/
```

4. Install the CAA Binary

```
sudo cp bin/caa /usr/local/bin/  
sudo chmod +x /usr/local/bin/caa
```

5. Configure Credentials

```
nano ~/.caa/caa.conf
```

Update the file:

```
Copernicus host: 192.168.254.1  
Username: YOUR_USERNAME  
Password: YOUR_PASSWORD
```

Important:

- Ensure the host is set to **192.168.254.1**
- Replace with your PES credentials
- Remove any existing **Encrypted password** line before adding **Password**

6. Run & Test the Authentication Agent

```
caa
```

Expected result:

- No password prompt
- Successful authentication
- Internet access enabled

7. Troubleshooting

- **No internet after running CAA:** Reconnect Wi-Fi and run `caa` again.
- **Authentication fails:** Verify username and password.
- **Permission denied:**

```
chmod +x /usr/local/bin/caa
```

- **Configuration missing:** Re-extract the package and copy `.caa` again.

Summary

1. Download client from Sophos portal
2. Extract package
3. Copy configuration files
4. Install binary
5. Configure credentials
6. Run CAA

Daily Usage & Important Notes

When Wi-Fi drops or you switch networks:

If CAA is running and you lose connectivity, press **Ctrl+C** in the terminal to stop it. Once you are reconnected to the Wi-Fi, simply run **caa** again to re-authenticate.

Do not log in through the browser captive portal.

Avoid logging into **rr.pes.edu:8090** through your browser for day-to-day internet access. Doing so may interfere with CAA and cause authentication to fail. Always use the **caa** command exclusively.

Why Use CAA?

Using CAA over the browser portal comes with several practical advantages:

- **No more max login limit errors** — CAA handles session management cleanly, so you will never hit the simultaneous login cap that the browser portal enforces.
- **No manual logout needed** — Switching devices or reconnecting does not require you to log out first. Just run **caa** and you are authenticated.
- **One command, every time** — Once configured, getting online is as simple as typing **caa** in a terminal. No browser, no portal, no hassle.
- **Works across reconnects** — Whether you wake your laptop from sleep or move between access points, a quick **caa** restores your session instantly.